

BUSINESS CONTEXT

This report presents a comprehensive analysis of multi-country retail sales performance across:

- 6 Countries (UK, Canada, US, India, China, Nigeria)
- 18 Store Locations
- Multiple Product Categories
- Customer Demographics
- Payment Channels

The analysis integrates:

- Enterprise Data Architecture (RAW → VALIDATION → TRANSFORM → FACT)
- SQL-based Exploratory Data Analysis
- Advanced Statistical Testing
- Multiple Regression Modeling
- Power BI Executive Dashboards

The objective was to move from:

“What happened?” → “Why did it happen?” → “What should management do?”

DATA GOVERNANCE & QUALITY ASSURANCE

Issue Identified

Initial dataset contained:

- Unrealistic cost-price structure
- Excessive discount rates (up to 98%)
- Artificial negative profit distortions

Corrective Actions Taken

- Reconstructed cost price (65–75% of selling price)
- Controlled discount distribution (0–40%)
- Recalculated:
 - Gross Revenue
 - Net Revenue
 - Total Cost
 - Total Profit

- Profit Margin

Result

- Financial logic restored
 - Profitability stabilized
 - KPI outputs now business-realistic
 - Dashboard insights reliable for decision-making
-

EXECUTIVE SUMMARY (High-Level KPIs)

From your Executive Dashboard:

- **Total Sales:** ~4.14M
- **Total Profit:** ~959K
- **Average Profit Margin:** ~23%
- **Average Basket Size:** 5.47 items
- **MoM Growth:** Positive in year-end months

Business Health

- Profitability stable
 - Revenue diversified across countries
 - No heavy dependency on single geography
 - Strong seasonal pattern (December peak)
-

KEY STRATEGIC INSIGHTS

A) COUNTRY PERFORMANCE

- UK leads in total revenue.
- Canada and US closely follow.
- Nigeria lowest contributor but not loss-making.
- Revenue spread fairly balanced.

Executive Interpretation:

Business is globally diversified and not geographically dependent.

B) CATEGORY PERFORMANCE

- Home & Kitchen, Clothing, Electronics are top profit drivers.

- Profit distribution across categories is balanced.
- No category dominance risk.

Strategic Meaning:

Portfolio diversification is healthy.

C) CUSTOMER INSIGHTS

Age Group:

- Adults contribute highest revenue.
- Seniors close behind.
- Young slightly lower.

But:

- T-Test → No statistical difference in purchase quantity.
- Cohen's $d = 0.015$ → Practically zero effect.

👉 Age is not a volume driver.

Gender:

- Male and Female contribution almost equal.
 - Balanced demographic profitability.
-

Payment Method:

- Mobile strong in developed countries.
- Cash still relevant in Nigeria.
- Chi-Square $p = 0.107$ → No dependency on age.
- Cramer's $V = 0.038$ → Extremely weak association.

👉 Payment behavior not age-driven.

5 SEASONAL & OPERATIONAL PATTERNS

Seasonal Trend:

- December peak (festive demand).
- Mid-year slowdown (April–June).
- Q4 strong rebound.

Operational Insight:

Inventory and marketing must align with seasonality.

6 RISK & VARIABILITY ANALYSIS

Revenue:

- Coefficient of Variation $\approx 82\%$
- High revenue volatility.
- Many small transactions + few large ones.

Profit Margin:

- CV $\approx 42\%$
- Moderately stable margins.

Outliers:

- Only 6 extreme transactions (Z-score > 3)
- No systemic pricing risk

👉 Business performance largely stable.

7 STATISTICAL TESTING SUMMARY

◆ T-Test (Adult vs Senior)

$p = 0.728$

Cohen's $d = 0.015$

✗ No age-based quantity difference.

◆ ANOVA (Store Location → Profit Margin)

$p = 0.629$

$\text{Eta}^2 = 0.0048$ (0.48%)

✗ Geography does not drive margin.

◆ Chi-Square (Age × Payment)

$p = 0.107$

Cramer's $V = 0.038$

✗ No demographic dependency in payment behavior.

8 MULTIPLE REGRESSION FINDINGS (Most Critical Section)

Model Strength

$R^2 = 4.94\%$

Adjusted $R^2 = 4.59\%$

Meaning:

Profit margin is multi-factorial and complex.

Significant Predictors

Quantity Purchased

Coefficient = +0.697

$p < 0.000001$

Strongest driver of profit margin.

Discount Applied

Not statistically significant.

Discount does not meaningfully affect profit after controlling for other variables.

Price Per Unit

Small but statistically significant negative effect.

Higher price slightly reduces margin efficiency.

Category Dummies

Not significant.

Country Dummies

Not significant.

THE CORE DISCOVERY

After eliminating:

- Age
- Gender
- Geography
- Category
- Discount
- Payment behavior

Only one strong driver remains:

PROFIT IS VOLUME-DRIVEN

STRATEGIC RECOMMENDATIONS

What NOT to Do

- Do not restructure by city
 - Do not run aggressive discount campaigns
 - Do not segment heavily by age
 - Do not over-focus on category differences
-

What TO DO

Increase Basket Size

- Product bundling
- Combo offers
- Cross-sell engines

Focus on Volume Growth

- Target high-performing stores
- Upselling strategy
- Loyalty incentives

Optimize Inventory for Seasonality

- Stock ahead of Q4
- Boost marketing before December

Improve Operational Efficiency

Since 95% of margin variance is unexplained:

- Supply chain optimization
 - Cost control
 - Vendor negotiations
-

EXPECTED BUSINESS IMPACT

If volume strategy is implemented:

- +4–7% margin efficiency improvement
- Improved revenue stability

- Reduced operational leakage
 - Better ROI on marketing spend
 - Stronger forecasting reliability
-

🏢 FINAL EXECUTIVE CONCLUSION

This project successfully:

- Built enterprise-grade data architecture
- Cleaned unrealistic synthetic data
- Conducted structured SQL EDA
- Applied statistical elimination logic
- Performed regression modeling
- Delivered interactive Power BI dashboards
- Identified true business profit driver

Executive-Level Insight:

This is a volume-sensitive retail business where scaling transactions improves margin efficiency more than discounting or geographic targeting.